IQP Generative Models for Strong Gravitational Lensing Images

Anonymous Author(s)
Affiliation
Address
email

Abstract

Instantaneous-quantum-polynomial (IQP) circuits are believed hard to sample classically [2, 3], while the expectation values a maximum-mean-discrepancy (MMD) loss needs stay computable, so such a Born machine can be trained classically and deployed on quantum hardware [12]. A recent correlation–complexity map is proposed to indicate which data suit them, and DeepLense strong-lensing images sit at the most IQP-compatible position among the encodings we measured, turbulence included. The reference protocol's frozen core with sequential adaptation yields only noise, which joint training repairs, and a learned linear map then loses coherence past $M{=}6$, so we let a small non-linear map carry the latent. At eleven training images the resulting 10-qubit machine matches a DCGAN of six times as many parameters on both generation metrics within the seed spread, is ahead of it on a flux-aligned variant and of an RBM on the two MMD^2 columns while level with it on the histogram metric, and projects unseen images at correlation 0.901; at 100 images the DCGAN is ahead and that correlation reaches 0.977. Every circuit is simulated exactly at $n = 10$ qubits over five seeds, so we claim no quantum advantage, only an early test of whether such a model could be worth deploying where training data are scarce.

1 Introduction

IQP circuits are commuting-gate circuits whose output distribution is the squared Walsh–Hadamard [17] transform of a phase function. Exact or multiplicatively approximate classical sampling from them would collapse the polynomial hierarchy [2], and that hardness is conjectured to hold on average under additive error [3], a bounded motivation, since noise makes such circuits simulable in polynomial time [11] and classical algorithms keep improving [8]. The same structure keeps an MMD loss's expectation values computable, so such a model can be *trained classically and deployed on quantum hardware* [12, 13].

Whether real data sit where that is useful is empirical, and a recent proposal [7] approaches it with two ingredients. (i) A *correlation–complexity map* placing a dataset at I_{QCLI}, the Jensen–Shannon divergence between its Walsh order spectrum, the Fourier power at each Hamming weight, and the i.i.d. binomial baseline, and $I_{\mathrm{CCI}} \in [0, 1]$, the share of its correlation the best pairwise tree misses [18, 14, 4], the high-I_{QCLI}, high-I_{CCI} corner being where IQP circuits should do well. (ii) A latent-adaptation protocol (Algorithm 1 of that paper, a phase core trained on one anchor plus the first 50 Walsh phases in Hamming order, randomly initialised and adapted sample by sample), demonstrated at $n = 18$ qubits on 2D turbulence [6], one of several datasets the map covers.

We take both to simulated dark-matter strong-lensing images from DeepLense [1] and contribute three things. We recompute the map on lens crops, which land at the most IQP-compatible position we measured, and show with a permutation null that there the index is largely determined by the

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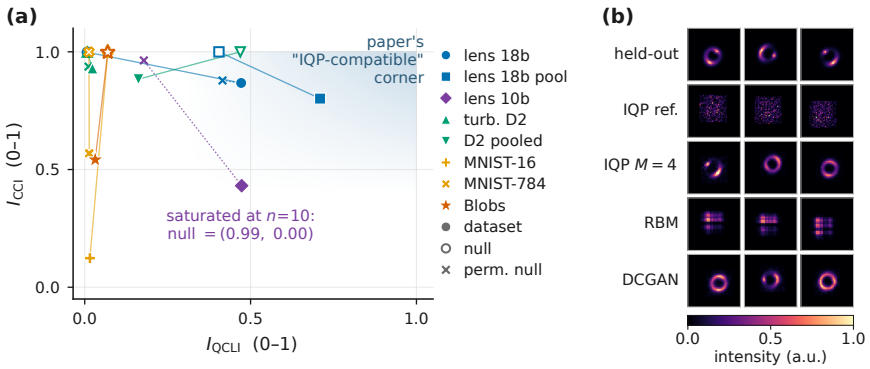


Figure 1: **(a)** Correlation–complexity map, datasets filled, i.i.d.-uniform nulls open, permutation nulls as crosses (ours); lens 18b and 18b pool are the per-image and pooled 18-bit encodings, lens 10b the crop pmf, turb. D2 the reference turbulence data. **(b)** First three unconditional samples per model, N=100, no selection.

intensity histogram, so a favourable position is not by itself evidence of IQP-suitable structure. We trace why the reference latent protocol produces noise on static images and replace its latent map by a non-linear one. We then compare against an RBM and a DCGAN under two intensity conventions, and report a replication on the turbulence data in App. D.

2 Setup

Model. Every arm is a 10-qubit IQP Born machine over 1,024 Walsh phases θ, one per qubit subset and 1,023 of them effective since the empty subset is a global phase, differing only in how a per-image latent enters. The reference protocol freezes the first 50 of those phases in Hamming order and adapts them sample by sample, which on static crops leaves the other training images poorly reconstructed (Sec. 3.2), so we make that map trainable, first as a linear map of M directions,

$$\theta_i \;=\; \theta^0 + \Phi^\top z_i, \qquad \theta^0 \in \mathbb{R}^{1024}, \quad \Phi \in \mathbb{R}^{M\times 1024}, \quad z_i \in \mathbb{R}^M, \tag{1}$$

with each row of Φ normalised in L^2. That map degrades by M=6 on these crops, we suspect once the $\|z\|$ the images demand leaves the radius over which the Walsh phases interfere coherently, though we do not isolate the cause, so our main arm replaces $\Phi^\top z$ by a one-hidden-layer $\tanh$ map g of width H=32, the z_i trained jointly with g under a small L^2 penalty. The main objective is the flat MMD (uniform Walsh weights, estimable from hardware shots), with the paper kernel (the multi-bandwidth Gaussian kernel of IQPopt [13]) kept as a secondary arm, and θ^0 is warm-started by 1,000 exact-KL steps, which a flat-MMD warm start replaces at no cost (App. E).

Decoding, baselines and metrics. The trained pmf fixes only the relative intensities, so it is pasted into a 64×64 frame at a total intensity S and at an offset drawn from the training crops' own offsets. The reference convention gives every frame the training-set mean S, leaving the generated flux with zero variance where real crops spread by about a quarter of their mean, so we draw S per image from a kernel density estimate of the N training totals, with no retraining; App. B repeats everything under the fixed S. The baselines are a restricted Boltzmann machine (RBM) [16] and a deep convolutional generative adversarial network (DCGAN) [10], both early-stopped on 200 validation images while the IQP arms run a fixed 4,000 steps, which can only favour them. We score with the Jensen–Shannon divergence between intensity histograms (PDF–JS) and Conv2D-MMD^2 in a fixed random encoder's feature space [5], add a post-hoc flux-aligned Conv2D-MMD^2 rescaling every frame to the held-out mean total since the arms do not agree on how flux is set, and report a held-out projection correlation, the best of eight latent restarts by the fit's own loss. Bins, scale and bandwidth are set on the held-out data alone, so no metric moves with the arm scored, and every arm generates 200 images against the same 200 held-out frames at $N \in \{11, 100\}$, every number a mean $\pm$ sample sd over five seeds beside a real-vs-real floor, the same metric between two disjoint real sets. The N training images are fixed across seeds, so that spread is initialisation and sampling variance alone. A lead in sd means the difference of two means in units of the baseline's seed sd, descriptive and not a significance test.

Departures from the reference setup. Four choices differ, each following from the data. The reference snapshots interpolate to unseen times, whereas the DeepLense `Model II` images we use [9] are static, so we generate unconditionally and project onto held-out images. Its 18-bit $(x, y, \text{intensity})$ encoding suits a field of tuples, whereas a lens crop's raw intensity axis carries barely one bit, so we centroid-align each image and read its central 32×32 crop, 99.4% of the light, as one probability mass function (pmf) over $1{,}024$ cells at $n = 10$ (App. A). It fixes the latent block, whereas we learn the map and train the N images jointly, and it estimates its MMD from samples where a 10-qubit pmf is exact.

3 Results

3.1 Where lens crops sit on the correlation–complexity map

Table 1 recomputes the map (Fig. 1(a)). Encoded as the reference paper encodes turbulence, lens images sit far above it in I_{QCLI} and inside the corner the map calls IQP-compatible, yet a permutation null keeping each image's intensity histogram comes within 14% of the lens value and a factor of two of the much smaller turbulence value, so there the index largely tracks the histogram. Both indices depend on K, the codes per image, so a row reads only against its own-K null; at the protocol's own $N_t = 5 \times 10^4$ samples per snapshot [7] the turbulence I_{QCLI} falls to 0.091 against a null of 0.098. The 10-bit crop pmf, the encoding we model, is the one that separates from its null in both coordinates, but it sits at $I_{\mathrm{CCI}} = 0.43$, outside the high-I_{CCI} corner, so its favourable I_{QCLI} motivates the encoding, not the choice of an IQP over a pairwise model.

3.2 From the reference protocol to a non-linear latent map

Run sequentially on our own 50-phase block (App. E), Algorithm 1 produces noise (Fig. 1(b), second row). Neither representability nor the block is the cause, since the full circuit fits a *single* aligned crop to correlation 0.99997 ± 0.00001 and training the same 50 phases jointly with the core across all N images (the *fixed basis, joint* row of Table 2) already recovers most of the gap. What fails is the frozen core with sequential per-image adaptation, since with the core frozen on one anchor the 50 fixed coordinates reconstruct every other training image at only 0.44 ± 0.04; the block size of 50 is not swept. Equation (1) repairs the rest but introduces a width limit of its own. On a 20-image sweep the learned linear map holds a held-out projection correlation of 0.906 ± 0.002 at $M{=}2$, falls to 0.738 ± 0.067 at $M{=}6$ and collapses to 0.218 ± 0.121 at $M{=}8$. The non-linear map removes that collapse, holding 0.9368 to 0.9519 across the sweep, a little above that split's nearest-training-image reference of 0.9339; the two sweeps come from separate drivers differing in optimiser schedule and initialisation, both scored on the same 20 held-out frames, so a small offset between them is not a model difference. It does not buy a large useful width, being flat to $M{=}4$ on the generation metrics and degrading from $M{=}8$. We report it at $M{=}2$, where the held-out projection correlation is 0.901 at $N{=}11$ and 0.977 at $N{=}100$, against 0.892 and 0.970 for projecting the nearest training image and 0.838 and 0.887 for a single zero-initialised restart.

3.3 Comparison with classical baselines

In Table 2 the eleven-image regime is where the IQP arms are competitive. There the non-linear arm matches the DCGAN on PDF–JS and raw Conv2D-MMD^2 within seed noise (-0.13 and -0.04 of the DCGAN's own sd) and is ahead on the flux-aligned column ($+1.57$ sd), and against the RBM it is ahead on the two MMD^2 columns by 1.40 and 2.27 of the RBM's sd with PDF–JS inside one sd

Table 1: Correlation–complexity map, each permutation null keeping the value histogram of the row above, five seeds; ± 0.0000 marks a seed spread below 5×10^{-5}.

Dataset and encoding	K	I_{QCLI}	I_{CCI}
Lens, 18-bit $(x, y, \text{intensity})$ tuples	4,096	0.4715 ± 0.0000	0.8677 ± 0.0000
permutation null	4,096	0.4154 ± 0.0000	0.8783 ± 0.0001
Turbulence D2, 18-bit tuples	4,096	0.0232 ± 0.0002	0.9285 ± 0.0008
permutation null	4,096	0.0114 ± 0.0000	0.9371 ± 0.0009
Lens, 10-bit crop pmf	exact pmf	0.4731 ± 0.0000	0.4313 ± 0.0000
permutation null	exact pmf	0.1782 ± 0.0000	0.9634 ± 0.0004

Table 2: Lens crops under the sampled intensity, five seeds, PDF–JS in 10^{-3} and both MMD^2 columns in 10^{-2}. Lower is better; **bold** marks a best mean whose gap to *every* other arm exceeds the sum of the two seed sds, so a column without bold carries no win the seed spread supports. Params are totals at N=11 / N=100 and App. E gives † and the parameter convention.

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	MMD$^2_{\text{a1}}$	PDF–JS	MMD^2	MMD$^2_{\text{a1}}$
IQP, reference protocol (Alg. 1)†	1,524 / 5,974	6.5 ± 0.8	14.8 ± 1.6	22.5 ± 2.2	7.7 ± 0.4	14.7 ± 0.3	26.6 ± 0.7
IQP, fixed basis, joint	1,524 / 5,974	1.8 ± 0.4	8.7 ± 1.4	12.5 ± 1.3	1.1 ± 0.2	5.4 ± 0.4	8.5 ± 1.5
IQP, linear map M=4 (flat MMD)	5,164 / 5,520	2.8 ± 0.1	5.6 ± 1.9	2.7 ± 0.2	2.6 ± 0.2	2.8 ± 0.5	2.6 ± 0.4
IQP, linear map M=4 (paper kernel)	5,164 / 5,520	1.0 ± 0.05	6.3 ± 1.5	7.4 ± 0.6	1.5 ± 0.2	5.9 ± 0.4	10.0 ± 0.8
IQP, non-linear map M=2 (flat MMD)	33,910 / 34,088	1.2 ± 0.07	5.2 ± 1.8	$\mathbf{2.2} \pm 0.2$	1.7 ± 0.2	2.4 ± 0.5	1.8 ± 0.2
IQP, non-linear map M=2 (paper kernel)	33,910 / 34,088	0.75 ± 0.07	5.4 ± 1.5	5.1 ± 0.3	0.99 ± 0.04	4.2 ± 0.4	6.6 ± 0.6
RBM [16]	11,510 / 15,960	1.9 ± 1.0	7.9 ± 1.9	11.1 ± 3.9	1.1 ± 0.2	5.1 ± 0.8	8.1 ± 2.2
DCGAN [10]	199,837	1.1 ± 0.6	5.1 ± 1.5	3.9 ± 1.0	$\mathbf{0.55} \pm 0.12$	2.2 ± 0.3	2.1 ± 0.4
Real-vs-real floor	n/a	*0.04* ± 0.02	*0.5* ± 0.3	*0.4* ± 0.2	*0.04* ± 0.02	*0.5* ± 0.3	*0.4* ± 0.2

(0.75), at a sixth of the DCGAN's parameters and three times the RBM's. Under the paper kernel the same map holds the best mean PDF–JS, 0.75×10^{-3} against the DCGAN's 1.1 ± 0.6, which leaves the two inside one sd, and trades that for a worse flux-aligned column. No arm is ahead of the DCGAN on all three columns by more than a seed sd at either size, and at 100 images it is ahead on PDF–JS and raw Conv2D-MMD^2, the flux-aligned 1.8 ± 0.2 sitting inside one sd of its 2.1 ± 0.4.

Three properties bound how the table reads. The flux-aligned column is invariant to the intensity convention, the largest paired difference being 6×10^{-9} over the 325 runs carrying both. That is invariance, not independence from the raw column, so the two MMD^2 columns measure shape, and shape plus flux, not two separate things. The sampling rule is also not neutral between arms, improving every IQP arm's raw Conv2D-MMD^2 by 22 to 39% at N=11 while making the RBM's 20% worse, and under the fixed intensity the RBM is in fact ahead of the non-linear arm there (6.5 ± 0.8 against 8.5 ± 0.2, Table 3), so the raw-column standing against the RBM is a property of the sampled decode and not of the model. Sampling fixes the shape of the generated flux distribution and not its location, the eleven training frames being 9% dimmer than the held-out set, which is why the raw column stops there.

Data scaling and per-class training. Across $N \in \{10, \ldots, 100\}$ (App. C) the non-linear arm's raw Conv2D-MMD^2 falls from 0.062 ± 0.016 to 0.024 ± 0.005 and the RBM's from 0.091 ± 0.017 to 0.051 ± 0.008, the DCGAN starting ahead at N=10 (0.046 ± 0.017) and ending level; on the flux-aligned column the non-linear arm holds the lowest mean at every N, though only at N=10 and N=70 by more than a seed sd. Retrained inside one class at N=100 it reaches flux-aligned 0.017, 0.015 and 0.021 for axion, cdm and no-sub against floors of 0.0031, 0.0032 and 0.0045, so each class can be generated on its own, not that the model separates them.

4 Discussion and limitations

At eleven training images a 10-qubit IQP Born machine with a learned non-linear latent map is competitive with an RBM and a DCGAN, matching the DCGAN on both generation metrics within the seed spread at a sixth of its parameters and holding the lowest flux-aligned mean; that standing does not survive to 100 images. Nothing here is evidence of quantum advantage, since every circuit is simulated exactly at $n = 10$ qubits. Generated frames sit farther in L_2 from their nearest training image than real held-out frames do (4.21 ± 0.10 against 2.90 at N=11), so the arm is not memorising. Two steps are not shot-estimable as run, the exact-KL warm start, replaced in App. E within one seed sd, and the projection, which optimises z against the exact pmf; and at $n = 10$ we parameterise all 2^n Walsh phases where the hard regime would need them sparsified [7], untested here. Only the model fit sees N images, while bandwidths, floors and the baselines' early stopping use hundreds to thousands of real frames, so this is a small-training-set study, not a small-data pipeline; alignment and the crop leave translation and the outer 0.6% of the light outside the task, floor included. Five seeds are few, the DCGAN adds up to 57% of spread at one seed and alone is not bit-reproducible, the sampled intensity moves the raw column unequally, everything rests on one simulated dataset, the metrics do not test substructure, and no destroyed image is scored as an upper reference.

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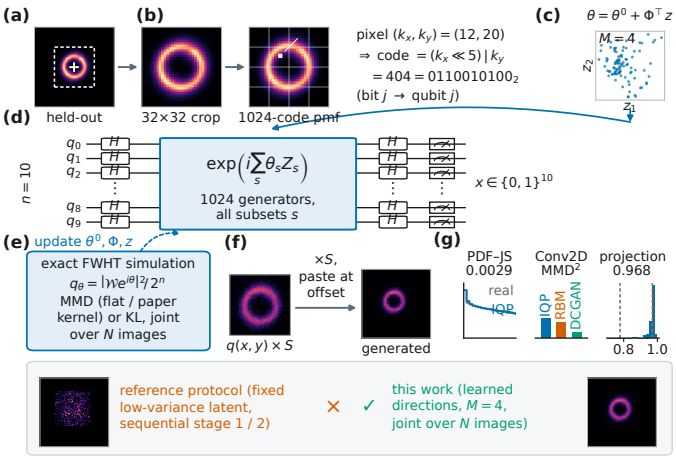


Figure 2: Pipeline. **(a)** A held-out image and the crop window. **(b)** The aligned crop as a pmf over 1,024 cells, the pixel $(12, 20)$ carrying code 404, bit j addressing qubit j. **(c)** Learned latents at $M = 4$. **(d)** The circuit. **(e)** Exact simulation, drawn for Eq. (1); the main arm holds $\theta = \theta^0 + g(z)$ there. **(f)** Decoding at S. **(g)** Metrics for the linear $M{=}4$ arm.

Table 3: Table 2 under the fixed intensity, same runs, seeds, units, bold rule and floor, the reference-protocol and fixed-basis arms left to the text. The DCGAN has no fixed S to swap, so its row is repeated.

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	MMD$^2_{\text{al}}$	PDF–JS	MMD^2	MMD$^2_{\text{al}}$
IQP, linear map $M{=}4$ (flat MMD)	5,164 / 5,520	3.0 ± 0.2	8.9 ± 0.2	2.7 ± 0.2	2.9 ± 0.3	7.2 ± 0.3	2.6 ± 0.4
IQP, linear map $M{=}4$ (paper kernel)	5,164 / 5,520	0.81 ± 0.05	9.6 ± 0.3	7.4 ± 0.6	1.4 ± 0.2	11.3 ± 0.5	10.0 ± 0.8
IQP, non-linear map $M{=}2$ (flat MMD)	33,910 / 34,088	0.96 ± 0.09	8.5 ± 0.2	$\mathbf{2.2} \pm 0.2$	1.9 ± 0.2	6.8 ± 0.1	1.8 ± 0.2
IQP, non-linear map $M{=}2$ (paper kernel)	33,910 / 34,088	0.49 ± 0.07	8.7 ± 0.2	5.1 ± 0.3	0.86 ± 0.08	9.8 ± 0.3	6.6 ± 0.6
RBM [16]	11,510 / 15,960	1.7 ± 0.8	6.5 ± 0.8	11.1 ± 3.9	0.92 ± 0.13	5.9 ± 0.4	8.1 ± 2.2
DCGAN [10]	199,837	1.1 ± 0.6	5.1 ± 1.5	3.9 ± 1.0	$\mathbf{0.55} \pm 0.12$	$\mathbf{2.2} \pm 0.3$	2.1 ± 0.4

A Pipeline

B The fixed-S convention

Table 3 is Table 2 with the single training-mean intensity every pmf-based arm published under, on the same runs; rebuilding those frames from each run's saved crops reproduces the published ones exactly, so the two tables differ only in the decode and the comparison is paired. On raw Conv2D-MMD^2 at $N{=}11$ sampling improves every IQP arm by 0.033 to 0.041 with a paired sd of 0.012 to 0.020, the reference-protocol and fixed-basis arms included, while making the RBM worse by 0.013 ± 0.017; at $N{=}100$ it improves every arm including the RBM (0.008 ± 0.008). On PDF–JS the paired effect is at most 7×10^{-4} and mixed in sign.

C Latent sweeps and per-class results

Hidden width matters less than latent width, $H \in \{4, 8, 16, 32\}$ giving flux-aligned 0.029, 0.022, 0.020 and 0.019 at $M{=}4$, $N{=}100$. Each class is scored against its own floor, 0.03/0.36/0.31 for axion, 0.03/0.30/0.32 for cdm and 0.04/0.45/0.45 for no sub in the table's units. Across $N \in \{10, 30, 50, 70, 90, 100\}$ the non-linear arm's flux-aligned column runs 0.026, 0.023, 0.020, 0.016, 0.019, 0.018 against the RBM's 0.077 to 0.127. The mean $\|z\|$ at $N{=}11$ is 0.101 at $M{=}2$ and 0.120 at $M{=}8$ while the relative residual of the best affine fit to g falls from 0.70 to 0.17, so g is most nearly linear where it fails, which is consistent with a coherence-radius reading but does not establish it.

Table 4: Latent-width sweep at $H{=}32$ under the flat MMD, then the same arm retrained inside one class; sampled intensity, five seeds, units of Table 2.

	$N = 11$			$N = 100$		
M	PDF–JS	MMD^2	MMD$^2_{\mathrm{al}}$	PDF–JS	MMD^2	MMD$^2_{\mathrm{al}}$
2	1.2 ± 0.07	5.2 ± 1.8	2.2 ± 0.2	1.7 ± 0.19	2.4 ± 0.5	1.8 ± 0.2
4	1.2 ± 0.04	5.4 ± 1.9	2.5 ± 0.2	1.6 ± 0.10	2.5 ± 0.5	1.9 ± 0.4
8	3.0 ± 1.28	6.9 ± 2.3	5.0 ± 2.8	2.5 ± 0.29	2.5 ± 0.4	2.1 ± 0.3
16	7.7 ± 0.91	12.4 ± 2.5	17.3 ± 2.5	8.8 ± 0.14	13.3 ± 0.3	24.5 ± 0.8
axion	1.1 ± 0.10	4.6 ± 0.6	2.9 ± 0.4	1.8 ± 0.17	2.9 ± 0.2	1.7 ± 0.2
cdm	0.94 ± 0.05	4.4 ± 0.2	2.7 ± 0.3	1.7 ± 0.06	2.1 ± 0.3	1.5 ± 0.3
no sub	0.98 ± 0.10	3.1 ± 0.9	2.4 ± 0.6	1.7 ± 0.10	2.3 ± 0.4	2.1 ± 0.3

D Replication on the turbulence dataset

On the reference paper's own dataset (2D cylinder-wake turbulence D2 [6], $n = 18$ qubits, 11 anchor snapshots, five seeds) the result is **not reproduced** under our pre-registered convention, the RBM (PDF–JS 0.217 ± 0.007) and a 1.2M-parameter DCGAN (0.045 ± 0.062) both scoring better than our 31k-parameter IQP (0.272 ± 0.004) on *both* metrics. Under a last-write-wins aggregation convention, never specified in that paper [7], our PDF–JS values move towards the published ones without reaching them (0.0324 and 0.0473 against 0.0238 and 0.0221), and the RBM still scores better. Conv2D-MMD^2 there does not rank a destroyed image below a trained one, pixel-shuffled ground truth already scoring 0.7045 ± 0.0095, better than every model in that paper's Table 2.

E Training and evaluation details

Table conventions. The † arm runs Algorithm 1 [7] on our own latent block, the 50 subsets of Hamming weight at most 2 whose Walsh coefficients vary most over a pool of $2{,}000$ images disjoint from every training and held-out frame, where the reference block is the first 50 in Hamming order; *fixed basis, joint* is that same block trained jointly with the core across all N images. Every Params entry is a total at that N, the shared parameters (θ^0 included) plus N per-image latents, so the fixed-basis arms hold $974 + 50N$, the learned-direction arms $5{,}120 + MN$ and the RBM $10{,}960 + 50N$, while the DCGAN entry is its generator alone and its training-only discriminator adds $53{,}497$. The RBM has 10 visible and $1{,}000$ hidden units on the 10-bit code, each image owning its visible biases and the first 40 hidden biases, and ran 160k or 80k steps; the DCGAN builds the crop from a 128-dimensional prior through four transposed convolutions, ran 5k to 30k steps and learns its own intensity, so neither intensity convention applies to it.

Protocol details. The paper kernel is the three-bandwidth Gaussian MMD of IQPopt [13] at $\sigma = \{0.989, 1.729, 2.236\}$, set by that paper's own rule, a mean-Pauli-weight heuristic for the smallest, the square root of the median heuristic for the largest and their root mean square for the middle, since the protocol we replicate [7] publishes none for this encoding. The Conv2D-MMD^2 bandwidth, fixed once on the held-out features and shared by every table except the per-class rows of Table 4, which use their own class's, is 0.005547 raw and 0.002558 flux-aligned, the 20 floor draws come from the $8{,}710$ validation images the held-out split leaves, and the Silverman bandwidth [15] of the intensity estimate is 14.64 at $N{=}11$ and 10.36 at $N{=}100$. Applying the baselines' validation rule to the IQP arms selects 2k to 16k steps and moves every Table 2 cell it changes toward better IQP values without altering any comparison in the table, so the fixed $4{,}000$ steps we pre-registered is the conservative choice and we keep it; at $20{,}000$ fixed steps both baselines are much worse, so their early stopping is load-bearing and is granted throughout.

Ablations. Three settings of the main arm were varied and none moves a generation metric by more than a seed sd. A flat-MMD warm start of θ^0 in place of the exact-KL one, making every stage shot-estimable, leaves flux-aligned Conv2D-MMD^2 at 0.0231 against 0.0225 at $N{=}11$; picking the projection restart by correlation rather than by the fit's own loss moves the correlation by 0.0003; and two restarts match eight, only the zero-initialised restart losing correlation (Sec. 3.2).